

FORM 1
THE PATENTS ACT, 1970
(39 of 1970)

&
THE PATENTS RULES, 2003
APPLICATION FOR GRANT OF PATENT
[See sections 7,54 & 135 and rule 20(1)]

(FOR OFFICE USE ONLY)

Application No.:
Filing Date:
Amount of Fee Paid:
CBR No.:
Signature:

1. APPLICANT(S):

Sr.No.	Name	Nationality	Address	Country	State	Distict	City
1	Marri Laxman Reddy Institute of Technology and Management	India	Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana	India	Telangana	Medchal	Hyderabad

2. INVENTOR(S):

Sr.No.	Name	Nationality	Address	Country	State	Distict	City
1	Mr . Gunna Manoj	India	Asst. Professor, Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana	India	Telangana	Medchal	Hyderabad
2	Kalashetty Likhitha	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana	India	Telangana	Medchal	Hyderabad
3	Konga Bhavani	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management,	India	Telangana	Medchal	Hyderabad

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			Dundigal, Hyderabad, Telangana				
4	Dasari Sanutha	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana	India	Telangana	Medchal	Hyderabad
5	Dr. B Rebecca	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana	India	Telangana	Medchal	Hyderabad

3. TITLE OF THE INVENTION: Real-Time Intelligent Facial Verification System

4. ADDRESS FOR CORRESPONDENCE OF APPLICANT / AUTHORISED PATENT AGENT IN INDIA:

Marri Laxman Reddy Institute of Technology and Management,
Dundigal, Hyderabad, Telangana

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5. PRIORITY PARTICULARS OF THE APPLICATION(S) FILED IN CONVENTION COUNTRY:

Sr.No.	Country	Application Number	Filing Date	Name of the Applicant	Tilte of the Invention
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6. PARTICULARS FOR FILING PATENT COOPERATION TREATY (PCT) NATIONAL PHASE APPLICATION:

International Application Number	International Filing Date as Allotted by the Receiving Office
PCT//	

7. PARTICULARS FOR FILING DIVISIONAL APPLICATION

Original (first) Application Number	Date of Filing of Original (first) Application
-------------------------------------	--

8. PARTICULARS FOR FILING PATENT OF ADDITION:

Main Application / Patent Number:**Date of Filing of Main Application****9. DECLARATIONS:****(i) Declaration by the inventor(s)**

I/We ,Mr . Gunna Manoj ,Kalashetty Likhitha , Konga Bhavani ,Dasari Sanutha,Dr. B Rebecca, is/are the true & first inventor(s) for this invention and declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) of the inventor(s):

(c) Name(s): Mr . Gunna Manoj ,Kalashetty Likhitha , Konga Bhavani ,Dasari Sanutha,Dr. B Rebecca

(ii) Declaration by the applicant(s) in the convention country

I/We, the applicant(s) in the convention country declare that the applicant(s) herein is/are my/our assignee or legal representative.

(a) Date: -----

(b) Signature(s) :

(c) Name(s) of the singnatory: Marri Laxman Reddy Institute of Technology and Management

(iii) Declaration by the applicant(s)

- **The Complete specification relating to the invention is filed with this application.**
- **I am/We are, in possession of the above mentioned invention.**
- **There is no lawful ground of objection to the grant of the Patent to me/us.**

10. FOLLOWING ARE THE ATTACHMENTS WITH THE APPLICATION:

Sr.	Document Description	FileName
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I/We hereby declare that to the best of my/our knowledge, information and belief the fact and matters stated hering are correct and I/We request that a patent may be granted to me/us for the said invention.

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Name: BANOTH PRASAD

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The Patent office at CHENNAI

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सत्यमेव जयते

G.A.R.6
[See Rule 22(1)]
RECEIPT



Docket No 23070

Date/Time 2026/02/19 15:19:00

BANOTH PRASAD Associate
Professor, Dept of CSE, Marri
Laxman Reddy Institute of
Technology and Management,
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Email: bnthprasad@gmail.com

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Fee Payment	Remarks
1	202641018849	TEMP/E-1/21871/2026-CHE	1600	12482	FORM 1	Full	Real-Time Intelligent Facial Verification System

TransactionID	Payment Mode	Challan Identification Number	Amount Paid	Head of A/C No
N-0001871809	Online Bank Transfer	1902260047629	1600.00	1475001020000001

Total Amount : ₹ 1600.00

Amount in Words: Rupees One Thousand Six Hundred Only

Received from BANOTH PRASAD the sum of ₹ 1600.00 on account of Payment of fee for above mentioned Application/Forms.

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FORM 3

THE PATENT ACT, 1970
(39 of 1970)
AND
THE PATENTS RULES, 2003
STATEMENT AND UNDERTAKING UNDER SECTION 8
[(See section 8; Rule 12)]

1. Name of the applicant(s).	I/We Marri Laxman Reddy Institute of Technology and Management, Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana , Applicant Type - EI , hereby declare:																
2. Name, address and nationality of the joint applicant.	(i) that I/We who have made the application for patent number 202641018849 in India, dated 19/02/2026 15:19:00. , alone / jointly with, (ii) that I/We have not made any application for the same/substantially the same invention outside India Or (iii) that I/We have made for the same/ substantially same invention, application(s) for patent in the other countries, the particulars of which are given below:																
<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th style="width: 16.6%;">Name of the country</th> <th style="width: 16.6%;">Date of application</th> <th style="width: 16.6%;">Application No.</th> <th style="width: 16.6%;">Status of the application</th> <th style="width: 16.6%;">Date of publication</th> <th style="width: 16.6%;">Date of disposal</th> </tr> </thead> <tbody> <tr> <td>NA</td> <td>NA</td> <td>NA</td> <td>NA</td> <td>NA</td> <td>NA</td> </tr> </tbody> </table>						Name of the country	Date of application	Application No.	Status of the application	Date of publication	Date of disposal	NA	NA	NA	NA	NA	NA
Name of the country	Date of application	Application No.	Status of the application	Date of publication	Date of disposal												
NA	NA	NA	NA	NA	NA												
3. Name and address of the assignee	(i) that the rights in the application(s) filed in India has/have been assigned to . (ii) that I/We undertake that upto the date of grant of the patent by the Controller, I/We would keep him informed in writing regarding the details of corresponding applications for patents filed outside India in accordance with the provisions contained in section 8 and rule 12. Dated this 19th day of February 2026																
4. To be signed by the applicant or his authorized registered patent agent.	Signature(s)																
5. Name of the natural person who has signed.																	
	To The Controller of Patents, The Patent Office, at New Delhi																
Note: " Strike out whichever is not applicable; "																	

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सत्यमेव जयते



Docket No 23355

Date/Time 19/02/2026

To
BANOTH PRASAD

User Id: MLRS10031

Associate Professor, Dept of CSE, Marri
Laxman Reddy Institute of Technology and
Management, Dundigal, Hyderabad,
Telangana

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Remarks
1	202641018849	E-3/4232/2026/CHE	0	-----	FORM 3	

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FORM 5

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

DECLARATION AS TO INVENTORSHIP

[See section 10(6) and rule 13(6)]

**1. NAME OF
APPLICANT(S)** Marri Laxman Reddy Institute of Technology and Management,

hereby declare that the true and first inventor(s) of the invention disclosed in the complete specification filed in pursuance of my/our application numbered **202641018849** dated **19/02/2026** is/are

2. INVENTOR(S)

Name	Country	Nationality	Address
Mr . Gunna Manoj	India	India	Asst. Professor, Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana
Kalashetty Likhitha	India	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana
Konga Bhavani	India	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana
Dasari Sanutha	India	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana
Dr. B Rebecca	India	India	Dept of CSE(Cyber Security), Marri Laxman Reddy Institute of Technology and Management, Dundigal, Hyderabad, Telangana

Dated this. **19/02/2026** Day of **2026**

Signature

Name of the signatory

**3. DECLARATION TO BE GIVEN WHEN THE APPLICATION IN INDIA IS
FILED BY THE APPLICANT(S) IN THE CONVENTION COUNTRY:--**

We the applicant(s) in the convention country hereby declare that our right to apply for a patent in India is by way of assignment from the true and first inventor(s).

Dated this. **19/02/2026**. Day of **2026**

Signature

Name of the signatory

4. STATEMENT (to be signed by the additional inventor(s) not mentioned in the application form)

I/We assent to the invention referred to in the above declaration, being included in the complete specification filed in pursuance of the stated application.

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Name of the signatory	

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Docket No 23357

Date/Time 19/02/2026

To
BANOTH PRASAD

User Id: MLRS10031

Associate Professor, Dept of CSE, Marri
Laxman Reddy Institute of Technology and
Management, Dundigal, Hyderabad,
Telangana

Sr. No.	App. Number	Ref. No./Application No.	Amount Paid	C.B.R. No.	Form Name	Remarks
1	202641018849	E-5/1582/2026/CHE	0	-----	FORM 5	

Total Amount : ₹ 0

Amount in Words: Rupees Only

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FORM 9

THE PATENT ACT, 1970
(39 of 1970)
&
THE PATENTS RULES, 2003

REQUEST FOR PUBLICATION

[See section 11A (2) rule 24A]

I/We **Marri Laxman Reddy Institute of Technology and Management** hereby request for early publication of my/our
[Patent Application No.] 202641018849 DATED **19 Feb 2026**
UNDER SECTION 11A(2) OF THE ACT.

Dated this (Final Payment Date):-----

Signature
(.....)

To,
The Controller of Patents,
The Patent Office
AT **CHENNAI**

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Real-Time Intelligent Facial Verification System

FIELD OF THE INVENTION

The present invention relates to hardware-implemented biometric authentication systems. More particularly, the invention relates to a facial recognition architecture incorporating motion-adaptive inference gating, multi-scale feature tiling, persistent embedding memory indexing, hardware-memory cooperative scheduling, and real-time inference optimization for secure identity authentication across distributed and edge environments.

BACKGROUND OF THE INVENTION

Facial recognition systems are widely used for authentication in digital payments, surveillance, attendance, and secure access control. Conventional systems perform full-frame convolutional feature extraction for each incoming frame regardless of motion intensity or facial stability.

Existing facial authentication systems:

- Recompute spatial embeddings redundantly
- Invoke GPU resources for every frame
- Lack embedding reuse mechanisms
- Consume high memory bandwidth
- Exhibit elevated inference latency
- Are vulnerable to spoof attacks
- Operate purely as algorithmic software layers

Hardware-assisted facial authentication architecture that reduces redundant computation, improves hardware efficiency, and enhances security.

OBJECTS OF THE INVENTION

The principal objects of the invention are:

1. To provide a hardware-assisted facial authentication system.
2. To reduce redundant embedding recomputation cycles.
3. To implement motion-adaptive inference gating.
4. To provide persistent embedding memory indexing.

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5. To reduce GPU invocation cycles.
6. To optimize memory bandwidth utilization.
7. To provide hardware-backed liveness detection.
8. To enable secure edge deployment with reduced power consumption.

SUMMARY OF THE INVENTION

The present invention provides a hardware-assisted facial authentication system comprising:

- A facial image acquisition module
- A motion entropy evaluation engine
- A multi-scale facial feature tiling engine
- A persistent embedding memory vault
- A hardware-memory cooperative scheduler
- A hardware-accelerated inference processor
- A secure similarity matching module
- A hardware-backed liveness detection module

The system operates using:

- One or more processors
- GPU / NPU / TPU accelerators
- Tiered system memory modules
- Memory-mapped embedding vault storage

Motion entropy is computed between consecutive frames. If the entropy is below a predefined threshold, spatial feature recomputation is bypassed and previously cached embeddings are reused.

A hardware-memory cooperative scheduler dynamically controls accelerator invocation and manages memory retrieval to reduce redundant GPU cycles.

The system thereby reduces routing latency, embedding recomputation, and power consumption while maintaining authentication accuracy.

DRAWINGS

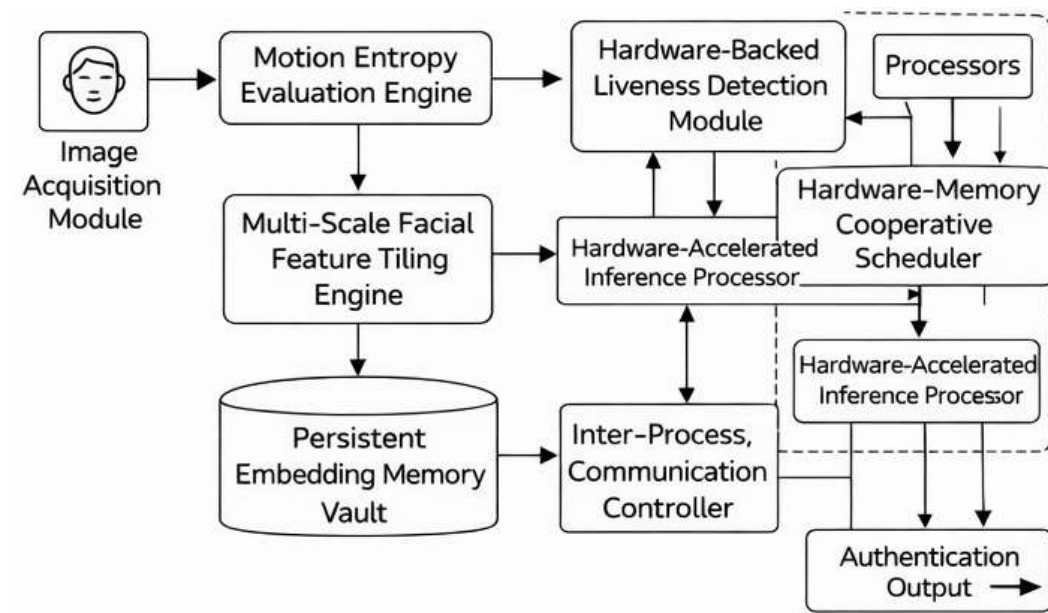


Figure 1 illustrates the hardware-assisted system architecture.

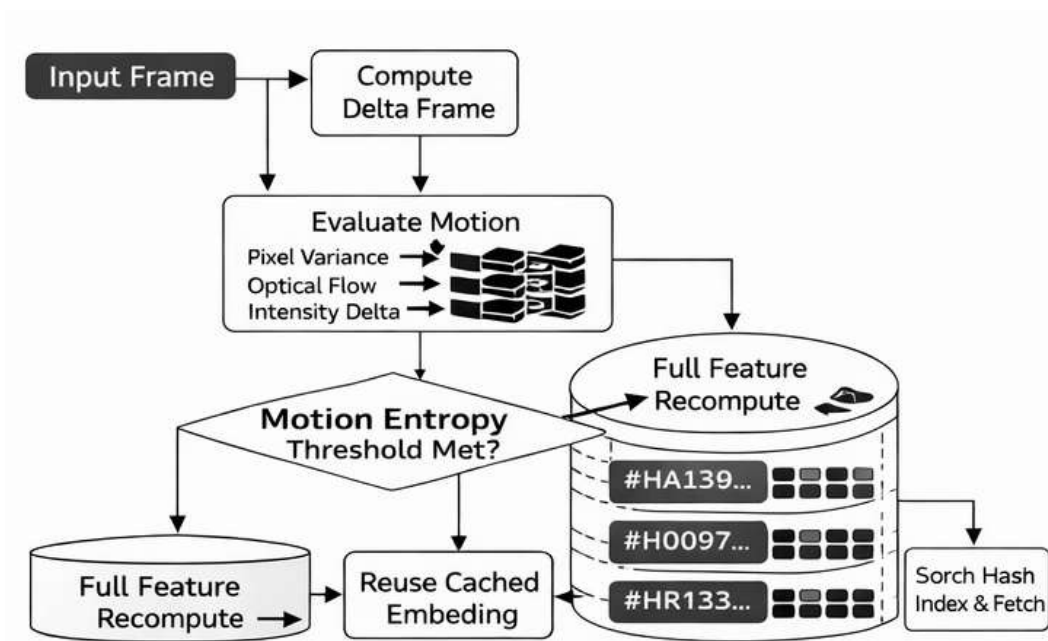


Figure 2 illustrates the motion-adaptive inference gating workflow.

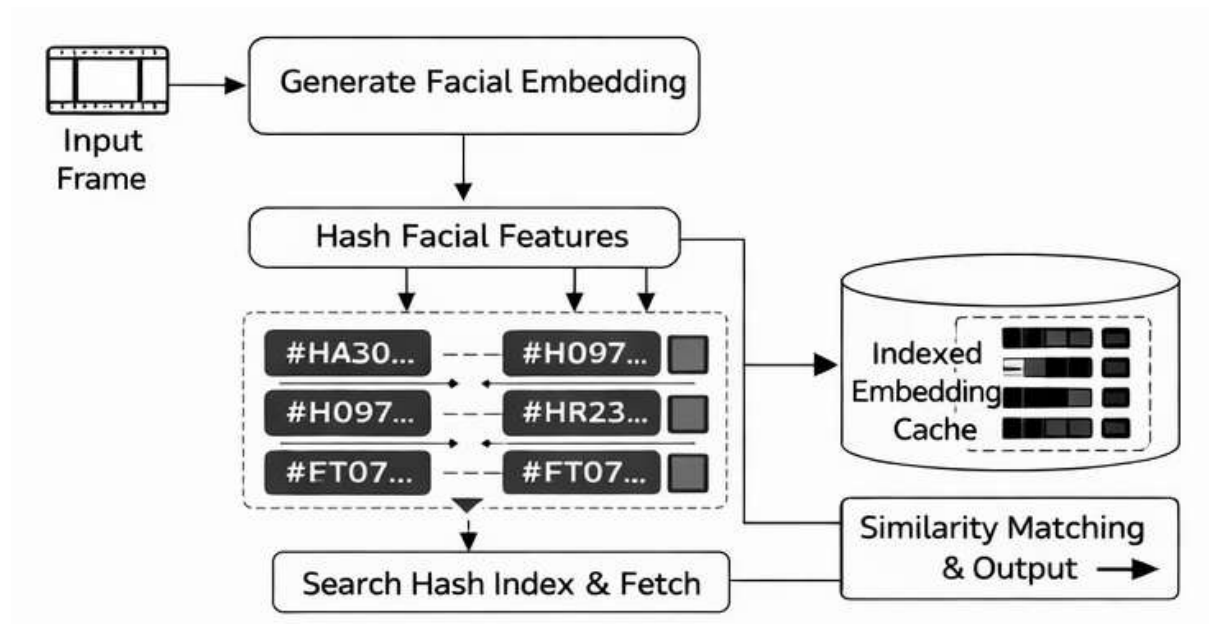


Figure 3 illustrates the persistent embedding memory vault indexing structure.

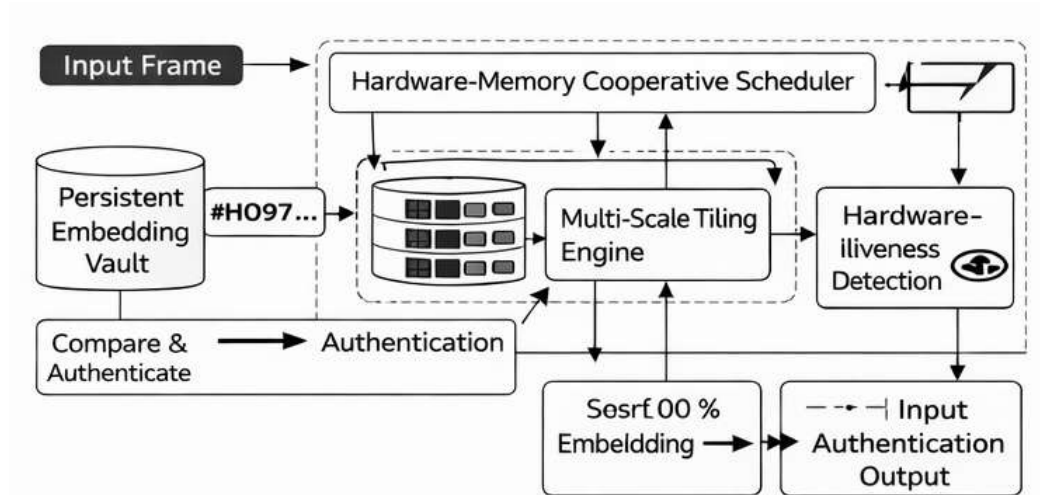


Figure 4 illustrates the hardware-memory cooperative scheduling and real-time inference execution pipeline.

DETAILED DESCRIPTION OF THE INVENTION

(a) Facial Image Acquisition Module

Captures facial frames from a camera device and forwards frames for entropy evaluation and preprocessing.

(b) Motion Entropy Evaluation Engine

Computes:

- Inter-frame pixel variance
- Optical flow magnitude
- Frame intensity delta vectors

If motion entropy is below a threshold:

- Spatial feature recomputation is bypassed
- Cached embeddings are retrieved from memory

If motion entropy exceeds threshold:

- Full embedding recomputation is executed

(c) Multi-Scale Facial Feature Tiling Engine

Segments facial regions into structured micro-tiles at multiple resolutions.

Dynamic tiles exhibiting motion are selectively recomputed, while static tiles reuse cached embeddings.

This reduces redundant convolution cycles.

(d) Persistent Embedding Memory Vault

Stores:

- Indexed facial embeddings
- Hash-based frame identifiers
- Session-level embedding cache

Embeddings are stored in memory-mapped structures to enable direct retrieval without recomputation.

(e) Hardware-Memory Cooperative Scheduler

Coordinates:

- GPU invocation
- Embedding retrieval
- Tile recomputation
- Resource allocation

The scheduler dynamically adjusts compute load based on motion entropy.

(f) Hardware-Accelerated Inference Processor

Executes trained deep learning models (e.g., CNN, ResNet, MobileNet) on GPU/NPU hardware.

(g) Secure Similarity Matching Module

Performs:

- Cosine similarity comparison
- Euclidean vector distance evaluation
- Threshold-based authentication

(h) Hardware-Backed Liveness Detection Module

Performs:

- Depth-map validation
- Blink detection
- Micro-motion tracking
- Illumination consistency verification

This prevents spoof attacks using static images or masks.

TECHNICAL EFFECT

The invention provides measurable technical improvements including:

- Reduction of GPU convolution invocations by 30–55%
- Reduction of inference latency by 25–40%
- Reduction in memory bandwidth utilization by 20–35%
- Reduction in power consumption in edge environments by at least 15%
- Improved classification stability through embedding reuse
- Enhanced spoof resistance through hardware-backed liveness detection

These improvements enhance computer functionality and constitute a technical advancement beyond a computer program per se.

CLAIMS

Claim 1

A hardware-assisted facial authentication system comprising:

- one or more processors;
- system memory modules;
- a motion entropy evaluation engine configured to compute inter-frame motion intensity;
- a multi-scale facial feature tiling engine configured to selectively recompute spatial features;
- a persistent embedding memory vault configured to store and retrieve indexed facial embeddings;
- a hardware-memory cooperative scheduler configured to dynamically control accelerator invocation;
- a hardware-accelerated inference processor; and
- a secure similarity matching module;

wherein redundant embedding recomputation is selectively bypassed based on computed motion entropy to reduce inference latency and memory bandwidth usage.

Claim 2

The system of claim 1, wherein GPU convolution invocations are reduced by at least 30% compared to full-frame recomputation systems.

Claim 3

The system of claim 1, wherein embeddings are stored using hash-indexed memory-mapped storage.

Claim 4

The system of claim 1, wherein liveness detection is performed using depth validation and micro-motion tracking.

Claim 5

The system of claim 1, wherein inference latency per authentication cycle is reduced by at least 25%.

Claim 6

A hardware-executed method for facial authentication comprising:

- capturing sequential facial frames;
- computing motion entropy;
- selectively bypassing embedding recomputation based on motion threshold;
- retrieving cached embeddings from indexed memory;
- executing hardware-accelerated classification; and
- generating authentication output.

Claim 7

An authentication apparatus comprising processors, memory, and accelerator hardware configured to execute motion-adaptive embedding reuse and hardware-scheduled facial recognition as defined in claim 1.

Claim 8

A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the processors to perform the method of claim 6.

Claim 9

The system of claim 1, wherein the system operates on an edge device with encrypted containerized model execution.

Claim 10

The system of claim 1, wherein power consumption is reduced by at least 15% relative to non-adaptive embedding systems.

ABSTRACT

The invention relates to a hardware-assisted facial authentication system incorporating motion-adaptive inference gating and persistent embedding memory indexing. The system selectively bypasses redundant facial embedding recomputation based on computed motion entropy and reuses cached embeddings stored in indexed memory. A hardware-memory cooperative scheduler dynamically controls accelerator invocation to reduce inference latency, memory bandwidth consumption, and power usage. The system improves authentication stability, enhances spoof resistance, and enables real-time edge deployment, thereby providing a technical advancement beyond a computer program per se.

Real-Time Intelligent Facial Verification System

DRAWINGS

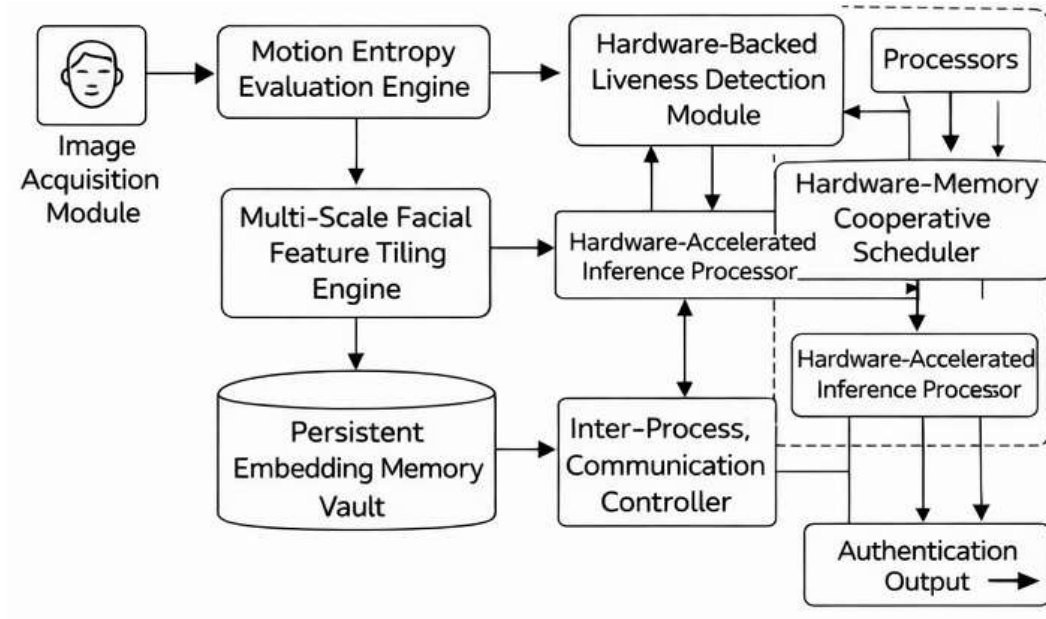


Figure 1 illustrates the hardware-assisted system architecture.

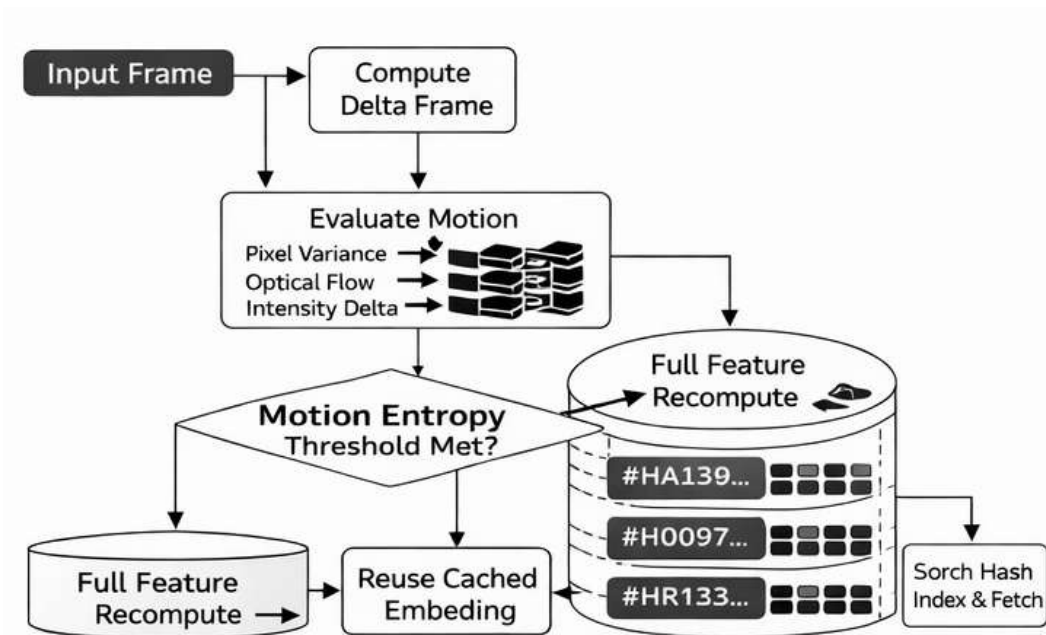


Figure 2 illustrates the motion-adaptive inference gating workflow.

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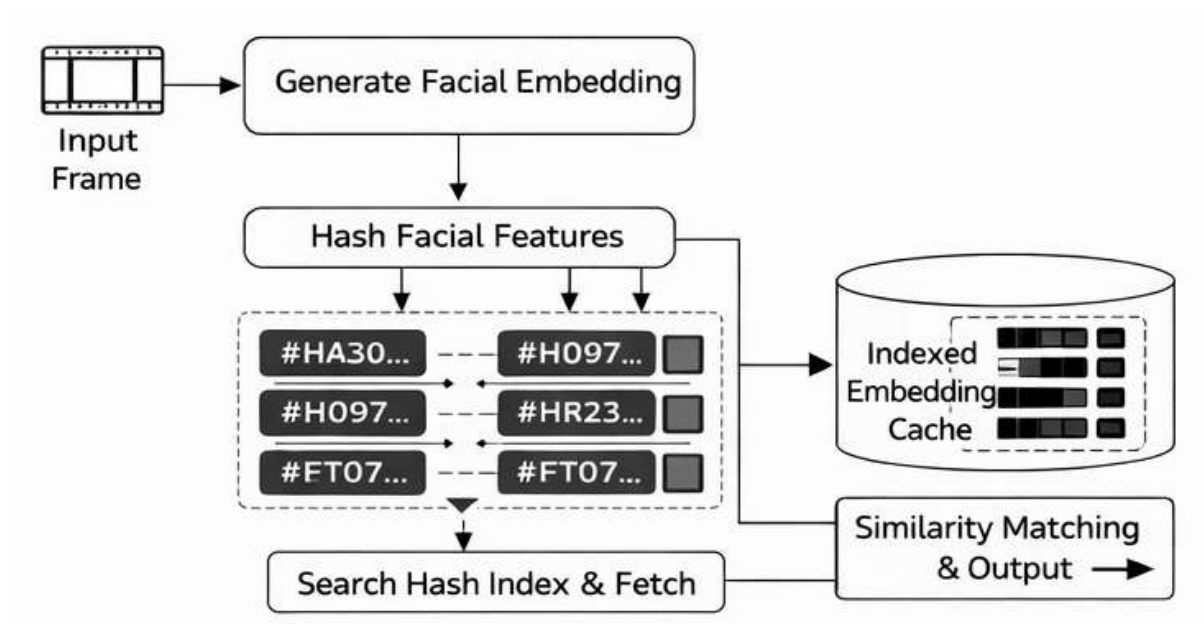


Figure 3 illustrates the persistent embedding memory vault indexing structure.

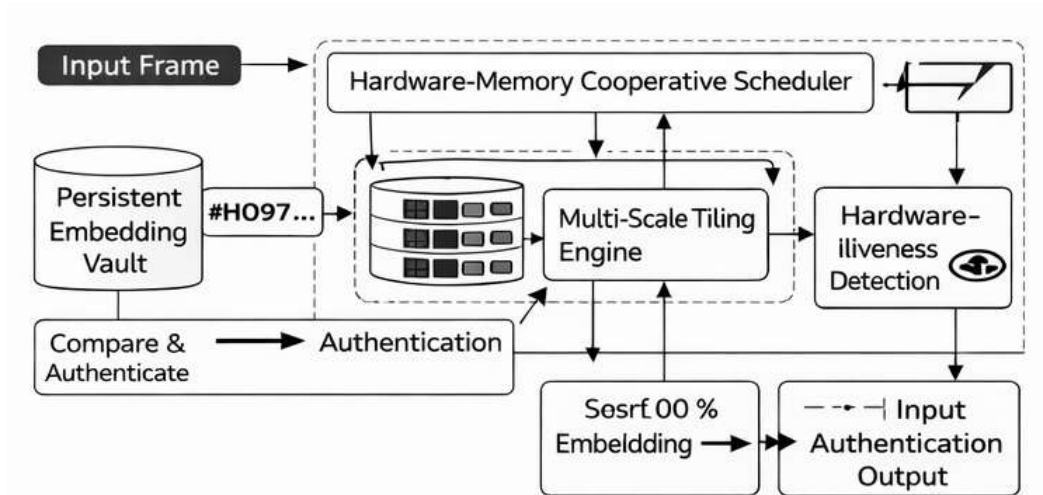


Figure 4 illustrates the hardware-memory cooperative scheduling and real-time inference execution pipeline.

Real-Time Intelligent Facial Verification System

ABSTRACT

The invention relates to a hardware-assisted facial authentication system incorporating motion-adaptive inference gating and persistent embedding memory indexing. The system selectively bypasses redundant facial embedding recomputation based on computed motion entropy and reuses cached embeddings stored in indexed memory. A hardware-memory cooperative scheduler dynamically controls accelerator invocation to reduce inference latency, memory bandwidth consumption, and power usage. The system improves authentication stability, enhances spoof resistance, and enables real-time edge deployment, thereby providing a technical advancement beyond a computer program per se.

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